

RESEARCH ARTICLE

Skeletal muscle bioenergetics during all-out exercise: mechanistic insight into the oxygen uptake slow component and neuromuscular fatigue

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Broxterman RM, Layec G, Hureau TJ, Amann M, Richardson RS. Skeletal muscle bioenergetics during all-out exercise: mechanistic insight into the oxygen uptake slow component and neuromuscular fatigue. *J Appl Physiol* 122: 1208–1217, 2017. First published February 16, 2017; doi:10.1152/jappphysiol.01093.2016.—Although all-out exercise protocols are commonly used, the physiological mechanisms underlying all-out exercise performance are still unclear, and an in-depth assessment of skeletal muscle bioenergetics is lacking. Therefore, phosphorus magnetic resonance spectroscopy (³¹P-MRS) was utilized to assess skeletal muscle bioenergetics during a 5-min all-out intermittent isometric knee-extensor protocol in eight healthy men. Metabolic perturbation, adenosine triphosphate (ATP) synthesis rates, ATP cost of contraction, and mitochondrial capacity were determined from intramuscular concentrations of phosphocreatine (PCr), inorganic phosphate (Pi), diprotonated phosphate (H₂PO₄[−]), and pH. Peripheral fatigue was determined by exercise-induced alterations in potentiated quadriceps twitch force (Q_{tw}) evoked by supramaximal electrical femoral nerve stimulation. The oxidative ATP synthesis rate (ATP_{OX}) attained and then maintained peak values throughout the protocol, despite an ~63% decrease in quadriceps maximal force production. Thus ATP_{OX} normalized to force production (ATP_{OX} gain) significantly increased throughout the exercise (1st min: 0.02 ± 0.01, 5th min: 0.04 ± 0.01 mM·min^{−1}·N^{−1}), as did the ATP cost of contraction (1st min: 0.048 ± 0.019, 5th min: 0.052 ± 0.015 mM·min^{−1}·N^{−1}). Additionally, the pre- to postexercise change in Q_{tw} (−52 ± 26%) was significantly correlated with the exercise-induced change in intramuscular pH ($r = 0.75$) and H₂PO₄[−] concentration ($r = 0.77$). In conclusion, the all-out exercise protocol utilized in the present study elicited a “slow component-like” increase in intramuscular ATP_{OX} gain as well as a progressive increase in the phosphate cost of contraction. Furthermore, the development of peripheral fatigue was closely related to the perturbation of specific fatigue-inducing intramuscular factors (i.e., pH and H₂PO₄[−] concentration).

NEW & NOTEWORTHY The physiological mechanisms and skeletal muscle bioenergetics underlying all-out exercise performance are unclear. This study revealed an increase in oxidative ATP synthesis rate gain and the ATP cost of contraction during all-out exercise. Furthermore, peripheral fatigue was related to the perturbation in pH and deprotonated phosphate ion. These findings support the concept that the oxygen uptake slow component arises from within active skeletal muscle and that skeletal muscle force generating capacity is linked to the intramuscular metabolic milieu.

ATP synthesis; ATP cost; magnetic resonance spectroscopy; muscle metabolism; neuromuscular fatigue

ALL-OUT EXERCISE PROTOCOLS are increasingly being utilized as an integral component of experimental designs (7, 22, 56, 58). The appeal of these protocols stems from the potential to determine multiple important physiological parameters in a single testing session. Indeed, maximal oxygen uptake ($\dot{V}O_{2\max}$), a gold standard in human physiology (19, 48), is elicited during all-out cycling exercise lasting longer than 90 s (7, 58). Furthermore, the 3-min all-out cycling protocol allows for the determination of critical power (CP) (for review see Refs. 7, 53), a parameter with great utility in both health and disease (25, 44, 55). This protocol has been adapted to provide equivalent parameters for other exercise modalities (4, 6, 9, 42, 57), such as critical force (CF), during a 5-min all-out knee-extensor protocol (6). However, despite growing use of these protocols, the physiological mechanisms underlying performance during all-out exercise are still under investigation, and an in-depth assessment of skeletal muscle bioenergetics (i.e., sources and rates of ATP synthesis) is currently lacking.

Interestingly, all-out exercise exhibits an oxygen uptake ($\dot{V}O_2$) slow component (7, 18, 52, 56, 59). The $\dot{V}O_2$ slow component, originally described for constant-work-rate exercise above the lactate threshold, is manifest as additional $\dot{V}O_2$, predominantly arising from a loss of efficiency within the exercising skeletal muscle (43, 45, 47), and is closely related to exercise tolerance (40). During all-out exercise, the $\dot{V}O_2$ slow component likely arises as a result of all motor units being recruited at the onset of exercise and motor units dropping out as fatigue ensues (i.e., decreased force production), while $\dot{V}O_{2\max}$ is maintained (7, 18, 52, 56, 59). Importantly, these findings demonstrate that progressive motor unit recruitment is not requisite for the $\dot{V}O_2$ slow component and therefore provides an interesting approach to study the mechanisms by which the $\dot{V}O_2$ slow component originates. In particular, it remains to be determined whether the slow component during all-out exercise is still evident during small-muscle-mass exercise and actually arises from the exercising skeletal muscle.

Remarkably, the magnitudes of intramuscular metabolic perturbation during the 3-min all-out cycling and 5-min all-out knee-extensor protocols are very similar (8, 52). Specifically, intramuscular phosphocreatine concentration ([PCr]) decreased to

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~20–25% of resting concentrations and muscle pH decreased to ~6.7, while muscle inorganic phosphate concentration ($[P_i]$) and blood lactate concentration increased to ~500% and ~1,200% of resting concentrations, respectively (8, 52). For high-intensity cycling exercise, the magnitude of intramuscular perturbation ($[P_i]$ and pH) is closely related to peripheral fatigue (2). However, the robustness of this relationship across exercise modality and exercising muscle mass remains unknown. Of note, the 5-min all-out knee-extensor protocol both induces a high degree of peripheral fatigue [~50% reduction in potentiated quadriceps twitch force (Q_{tw})] (6) and allows for phosphorus magnetic resonance spectroscopy (^{31}P -MRS) measurements of intramuscular metabolites (8), making this an ideal exercise paradigm to further assess the relationship between the magnitude of intramuscular perturbation and the development of peripheral fatigue.

Therefore, the purpose of this study was to quantitatively characterize skeletal muscle bioenergetics and peripheral fatigue during all-out exercise. Specifically, we utilized ^{31}P -MRS to determine the rate and source of ATP production and electrical femoral nerve stimulation to determine peripheral fatigue for all-out intermittent isometric single-leg knee-extensor exercise. We hypothesized that during the all-out exercise 1) the oxidative ATP synthesis rate (ATP_{OX}) would reach and then remain at peak values, while force production would progressively decrease, indicating an intramuscular $\dot{V}\text{O}_2$ slow component; 2) the ATP cost of contraction would progressively increase; and 3) in terms of fatigue, the reduction in quadriceps Q_{tw} would be related to the change in both intramuscular $[P_i]$ and pH.

METHODS

Subjects. Eight healthy men (age 28 ± 5 yr, stature 178 ± 4 cm, and body mass 77 ± 8 kg) volunteered and provided written informed consent to participate in this investigation. All experimental procedures were conducted in accordance with the Declaration of Helsinki and were approved by the Institutional Review Boards of the University of Utah and the Salt Lake City Department of Veterans Affairs Medical Center. Subjects were instructed to abstain from vigorous activity during the 24 h preceding each visit to the laboratory and to arrive at the laboratory having abstained from food and caffeine during the preceding 3 h. Subjects were tested in the laboratory a minimum of twice with at least 72 h between visits.

Experimental design. Single-leg intermittent isometric knee-extensor exercise (3-s contraction and 2-s relaxation) was performed for all exercise protocols. This was conducted in a semirecumbent position (~15° elevation of the trunk), with the knee of the leg to be exercised situated over a custom-built knee support (~45° knee joint angle), the ankle fixed to an immovable strain gauge (SSM-AJ-250, Interface), and nonelastic straps positioned over the hips and thigh. The choice of exercising leg was balanced for dominance across subjects. During the initial visit, subjects performed 60 maximal voluntary quadriceps contractions (MVCs) over 5 min with neuromuscular function assessments before and immediately after exercise (i.e., at the 60th MVC). This 60-MVC protocol with 5 min of recovery was then completed inside a whole body magnetic resonance imaging (MRI) system during the second visit. An audio recording cued the start and stop of each 3-s contraction, but no information regarding the duration of exercise was provided to the subjects. The integrated force, mean force, and peak force were determined for each of the 60 MVCs, and CF was defined as the mean force of the final 6 MVCs (6).

Neuromuscular function. During the initial visit, neuromuscular function assessments were conducted before and immediately after

exercise. The femoral nerve was stimulated with a constant-current stimulator (model DS7AH, Digitimer), with the anode placed in the femoral triangle and the cathode placed between the greater trochanter and the iliac crest. Low-intensity single-pulse stimuli (200- μs pulse width, 100–150 mA) were used to locate the optimal position of the stimulating electrode, defined as the location evoking the greatest force production. The electrodes were then fixed in position until all measurements for the visit were conducted. The stimulation intensity was increased in 20-mA increments until maximal force was obtained. The stimulator intensity was then set to 120% of this, to ensure supramaximality. For the evaluation of quadriceps function, Q_{tw} was performed 2 s after 3-s MVCs both before exercise and then immediately after exercise (i.e., at the 60th MVC). For each Q_{tw} , the contraction time ($\text{CT}_{Q_{tw}}$) and half-relaxation time ($0.5\text{RT}_{Q_{tw}}$) were determined. During each MVC, a superimposed twitch was delivered and voluntary activation of the quadriceps was calculated as $\text{VA} = [1 - (\text{superimposed twitch}/Q_{tw})] \times 100$.

^{31}P -MRS. A clinical 2.9-T MRI system (Tim-Trio, Siemens Medical Systems, Munich, Germany) operating at 49.9 MHz (^{31}P resonance) and a dual-tuned ^{31}P - ^1H surface coil (110-mm ^1H coil loop surrounded by ^{31}P single-loop coil with a diameter of 125 mm) with linear polarization were utilized to acquire ^{31}P -MRS data (RAPID Biomedical, Rimpar, Germany). The surface coil was secured over the mid thigh with elastic straps, and advanced localized volume shimming was performed after a three-plane scout proton image was acquired to ensure that all major quadriceps muscles were sampled. Two fully relaxed spectra were acquired (3 averages per spectrum and a repetition time of 30 s) before the 60-MVC protocol commenced. Throughout exercise and recovery, MRS data acquisition was conducted with a free-induction decay pulse sequence with a 2.56-ms adiabatic-half-passage excitation RF pulse, a repetition time of 2.5 s, a receiver bandwidth of 5 kHz, 1,024 data points, and 2 averages per spectrum. Comparisons between fully relaxed and partially relaxed spectra were utilized to quantify saturation factors.

A time-domain fitting routine using the AMARES algorithm (51) incorporated into CSAIPO software (33–35) was utilized to determine absolute and relative concentrations of intramuscular PCr, P_i , H_2PO_4^- , and ATP. Intracellular pH was calculated from the chemical shift difference between the P_i and PCr signals. The free cytosolic [ADP] was calculated from [PCr] and pH with the creatine kinase (CK) equilibrium constant ($K_{\text{CK}} = 1.66 \times 10^9 \text{ M}^{-1}$) and the assumption that PCr represents 85% of the total creatine content (23). Resting concentrations were calculated from the average peak areas of the two fully relaxed spectra and assuming a resting [ATP] of 8.2 mM (16). To account for P_i splitting, the pH corresponding to each P_i pool was calculated separately as pH₁ and pH₂ on the basis of the chemical shift of each peak relative to PCr, such that the overall pH was then calculated as

$$\text{pH} = \text{pH}_1 \times (\text{area}P_{i1}/\text{total } P_i \text{ area}) + \text{pH}_2 \times (\text{area}P_{i2}/\text{total } P_i \text{ area})$$

The concentration of H_2PO_4^- was calculated as (32)

$$\text{H}_2\text{PO}_4^- = [P_i]/(1 + 10^{\text{pH}-6.75})$$

Free cytosolic adenosine monophosphate (AMP) was calculated based on the equilibrium of the adenylate kinase reaction corrected for the effects of pH and assuming a free magnesium concentration of 1 mM (15). Relative amplitudes were corrected for partial saturation due to the repetition time relative to T1 with the fully relaxed spectra acquired at rest.

ATP synthesis rates and ATP cost of contraction. The rate of ATP production from the breakdown of PCr through the CK reaction (ATP_{CK} , mM/min) was calculated from the change in [PCr] for each time point of the exercise period (27):

$$\text{ATP}_{\text{CK}} = \Delta\text{PCr}/\Delta t$$

Based on the sigmoid relationship between ATP_{OX} (mM/min) and free cytosolic [ADP], the rate of mitochondrial ATP production was calculated as

$$\text{ATP}_{\text{OX}} = V_{\text{max}} / (1 + (K_m / [\text{ADP}])^{2.2})$$

in which K_m (the [ADP] at half-maximal oxidation rate) is $\sim 30 \mu\text{M}$ in skeletal muscle (27), 2.2 is the Hill coefficient for a sigmoid function (24), and V_{max} is the peak rate of in vivo oxidative ATP synthesis (see *PCr recovery kinetics*).

During exercise, changes in intramuscular pH result from glycogen breakdown to pyruvate and lactate, proton efflux, buffering capacity, protons produced by oxidative phosphorylation, and the consumption of protons by the CK reaction (27). Assuming that the glycogenolytic production of 1 mol of H^+ , when coupled to ATP hydrolysis, yields 1.5 mol of ATP, the ATP production from anaerobic glycolysis (ATP_{GLY}) can be deduced from the total number of protons (P) produced throughout exercise (20, 27, 28):

$$\text{ATP}_{\text{GLY}} = 1.5 \times P$$

where

$$P = \text{H}_{\text{CK}}^+ + \text{H}_{\beta}^+ - \text{H}_{\text{OX}}^+ + \text{H}_{\text{efflux}}^+$$

H_{CK}^+ (in mM/min) was calculated from the time-dependent changes in [PCr] and from the stoichiometric coefficient (γ):

$$\text{H}_{\text{CK}}^+ = -\gamma \times \text{ATP}_{\text{CK}}$$

where γ is the proton stoichiometric coefficient of the coupled Lohmann reaction as previously described (31). H_{β}^+ (in mM/min) was calculated from the apparent buffering capacity β_{total} (in slykes, millimoles acid added/unit change in pH) and from the rate of pH changes:

$$\text{H}_{\beta}^+ = -\beta_{\text{total}} \times \Delta\text{pH}/\Delta t$$

where

$$\beta_{\text{total}} = \beta_{\text{nonbicarbonate-nonP}_i} + \beta_{\text{P}_i} + \beta_{\text{bicarbonate}}$$

where

$$\beta_{\text{nonbicarbonate-nonP}_i} = \beta_a - \beta_{\text{P}_i}$$

in which β_a was determined from the initial change in PCr (ΔPCr_i) and alkalization of pH (ΔpH) (10):

$$\beta_a = \gamma \times (\text{PCr}_i \times \Delta\text{pH})$$

β_{P_i} was determined based on the dissociation constant of the buffer (K) according to the standard formula (11):

$$\beta_{\text{P}_i} = (2.303 \times \text{H}^+ \times K \times [\text{P}_i]) / (K + \text{H}^+)^2$$

where $K = 1.77 \times 10^{-7}$. In agreement with previous studies and assuming that muscle is a closed system during exercise (11, 28), $\beta_{\text{bicarbonate}}$ was set to zero. H_{OX}^+ (in mM/min) was calculated from the factor $m = 0.16/[1 + 10^{(6.1 - \text{pH})}]$, which accounts for the amount of protons produced through oxidative ATP production (28, 29):

$$\text{H}_{\text{OX}}^+ = m \times \text{ATP}_{\text{OX}}$$

$\text{H}_{\text{efflux}}^+$ (in mM/min) was calculated for each time point of exercise using the proportionality constant λ relating proton efflux rate to ΔpH (28, 29):

$$\text{H}_{\text{efflux}}^+ = -\lambda \Delta\text{pH}$$

This proportionality constant λ (in $\text{mM} \cdot \text{min}^{-1} \cdot \text{pH unit}^{-1}$) was calculated during the recovery period:

$$\lambda = -V_{\text{efflux}}/\Delta\text{pH}$$

During the recovery period, PCr is regenerated throughout the CK reaction as the consequence of oxidative ATP production in mitochondria. Thus $\text{H}_{\text{efflux}}^+$ can be calculated from the rates of proton production from the CK reaction (H_{CK}^+ , in mM/min) and mitochondrial ATP production (H_{OX}^+ , in mM/min) on one side and the rate of pH changes on the other side. At this time, ATP production is exclusively aerobic and lactate production is considered negligible:

$$V_{\text{efflux}} = \beta_{\text{total}} \times \Delta\text{pH}/\Delta t + \gamma \times V_{\text{PCr}} + m \times \text{ATP}_{\text{OX}}$$

To improve precision, a modified version of this calculation was used (28, 29), in which the total proton disappearance (i.e., $\int \text{Edt}$) is estimated cumulatively from the start of recovery and then fitted to an exponential function to obtain the initial recovery rate (V_{efflux}).

The total ATPase rate ($\text{ATP}_{\text{TOTAL}}$, in mM/min) was calculated for each time point as

$$\text{ATP}_{\text{TOTAL}} = \text{ATP}_{\text{OX}} + \text{ATP}_{\text{CK}} + \text{ATP}_{\text{GLY}}$$

and the anaerobic ATPase rate (ATP_{ANA} , mM/min) was calculated for each time point as

$$\text{ATP}_{\text{ANA}} = \text{ATP}_{\text{CK}} + \text{ATP}_{\text{GLY}}$$

The ATP cost of contraction (in mM/N) was calculated as the ratio between $\text{ATP}_{\text{TOTAL}}$ and the force integral. The ATP_{OX} and ATP_{ANA} gains were determined as the respective ATP synthesis rate normalized to the force integral. All values were calculated for each minute of exercise.

PCr recovery kinetics. The [PCr] kinetics during the recovery period were described by a monoexponential curve:

$$[\text{PCr}](t) = [\text{PCr}]_{\text{end}} + \Delta[\text{PCr}] \times (1 - e^{-(t/\tau_p)})$$

where $[\text{PCr}](t)$ is the [PCr] at a given time t , $[\text{PCr}]_{\text{end}}$ is the [PCr] at end exercise, $\Delta[\text{PCr}]$ is the amount of PCr resynthesized during the recovery period, and τ represents the time constant of the PCr offset kinetics. The initial rate of PCr resynthesis (V_{PCr}) was calculated from the derivative of the monoexponential equation at the onset of recovery:

$$V_{\text{PCr}} = k \times \Delta[\text{PCr}]$$

where $\Delta[\text{PCr}]$ is the amount of PCr resynthesized during the recovery period and the rate constant $k = 1/\tau$ (27). The peak rate of in vivo oxidative ATP synthesis (V_{max}) was calculated with V_{PCr} and the [ADP] at end exercise (50):

$$V_{\text{max}} = V_{\text{PCr}} \times [1 + (K_m / [\text{ADP}]_{\text{end}})^{2.2}]$$

where K_m (the [ADP] at half the highest oxidative rate) is $\sim 30 \mu\text{M}$ in skeletal muscle (27).

Model variables were determined with an iterative process by minimizing the sum of squared residuals (RSS) between the fitted function and the observed values. Goodness of fit was assessed by visual inspection of the residual plot and the frequency plot distribution of the residuals, with the χ^2 values and the coefficient of determination (r^2) calculated as follows (39):

$$r^2 = 1 - (\text{SS}_{\text{reg}}/\text{SS}_{\text{tot}})$$

Statistical analysis. Force, ^{31}P -MRS, ATP synthesis, ATP cost of contraction, ATP_{OX} gain, and ATP_{ANA} gain were analyzed by one-way ANOVAs with repeated measures. A two-way ANOVA with repeated measures was used to analyze the percent contribution of ATP_{OX} and ATP_{ANA} to $\text{ATP}_{\text{TOTAL}}$. Tukey's post hoc analyses were conducted when significant main effects were detected. Preexercise-to-postexercise comparisons for MVC, Q_{tw} , and VA were made with Student's paired t -tests. ATP_{OX} was compared with the 95% confidence interval (CI) around V_{max} to determine whether maximal oxidative ATP production was attained. The relationships between neuromuscular function measurements and intramuscular metabolic

perturbations were assessed with Pearson product moment correlation coefficients. Significance for the statistical analysis was accepted at $P < 0.05$. Results are presented as means $\pm$ SD, except in figures, where SE is used for clarity.

RESULTS

Force production, intramuscular metabolites, and neuromuscular function. The peak MVC force attained during the protocol was 568 ± 126 N, which significantly decreased to 234 ± 50 N by the final MVC (Fig. 1). Throughout the majority of the exercise protocol both the peak force and the integrated force per MVC significantly decreased over time (Fig. 1). However, as illustrated, force did not significantly change over the final 30 s of the test and the resulting CF was 209 ± 41 N, which corresponded to $38 \pm 10\%$ of the peak force during the protocol. The intramuscular metabolic responses to the 60-MVC protocol and postexercise recovery are presented in Fig. 2. All variables changed significantly as a function of time during both the exercise and recovery periods. Intramuscular pH remained significantly below, and $[H^+]$ significantly above, baseline values for the entire duration of the

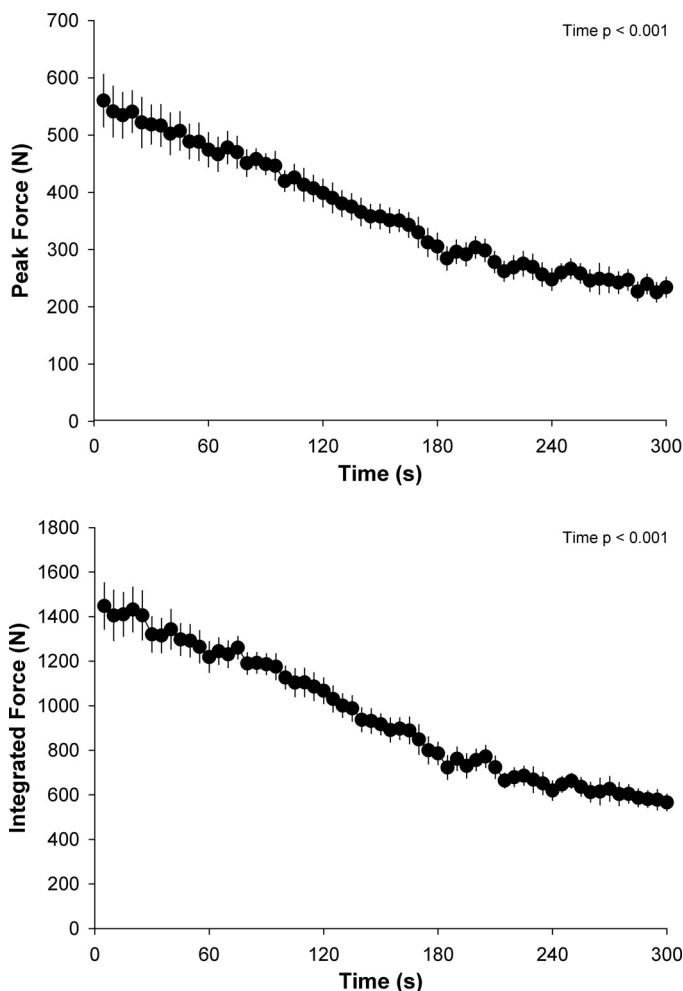


Fig. 1. Force development during the 5-min all-out intermittent isometric single-leg knee-extensor protocol. Subjects performed a series of 60 intermittent maximal voluntary contractions (3-s contraction, 2-s relaxation) over 5 min. Peak force and mean force were determined per maximal voluntary contraction.

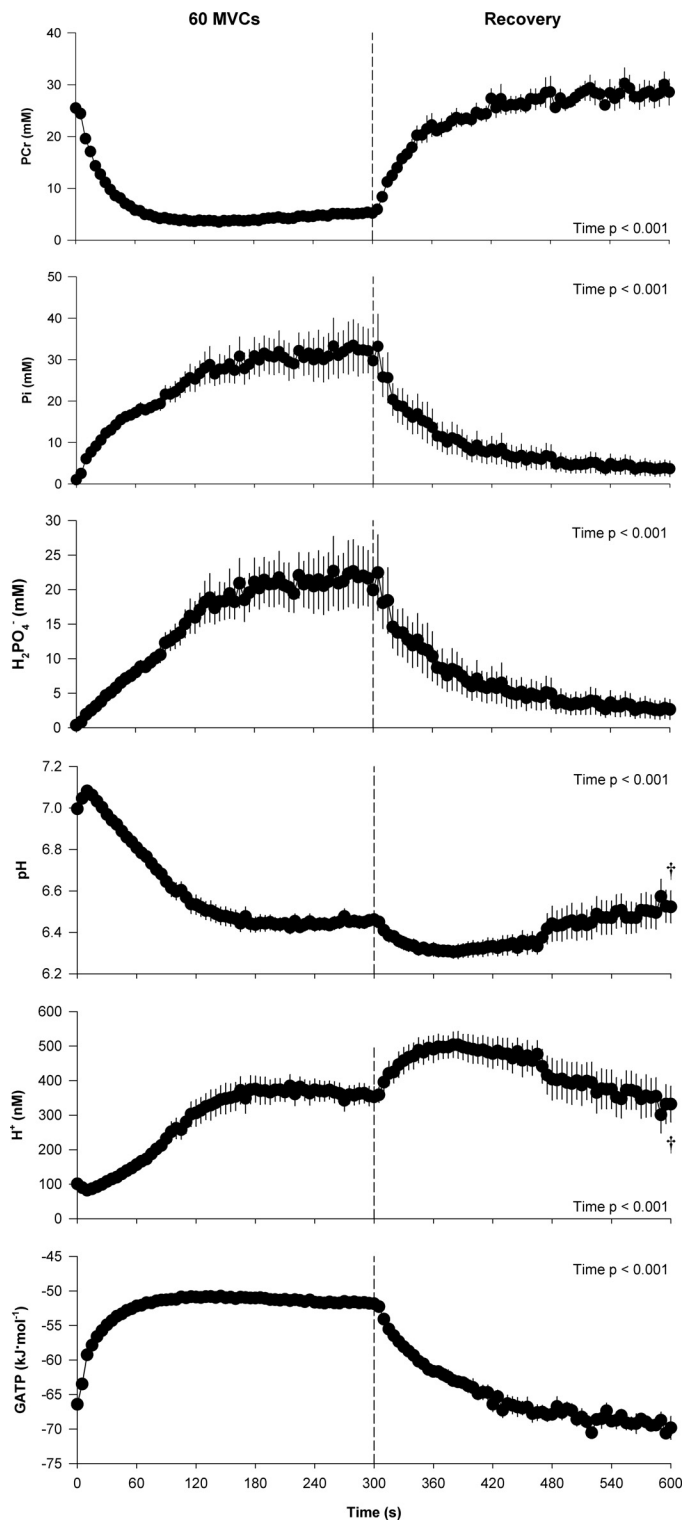


Fig. 2. Intramuscular metabolic perturbation during the 5-min all-out intermittent isometric single-leg knee-extensor protocol. Intramuscular metabolite concentrations were determined with phosphorus magnetic resonance spectroscopy. †Significantly different from baseline.

5-min recovery period, while $[PCr]$, $[P_i]$, $[H_2PO_4^-]$, and free energy of ATP hydrolysis $[\Delta G_{ATP}]$ were not significantly different from baseline values by the end of this period. The force generating capacity of the quadriceps was significantly

reduced immediately after exercise, as measured by the pre- to postexercise reduction in MVC. This decrease was accompanied by significant central (Δ VA: $-12 \pm 11\%$) and peripheral (Δ Q_{tw}: $-52 \pm 26\%$) fatigue (Table 1). The fall in Q_{tw} was significantly correlated with the decrease in intramuscular pH and increase in intramuscular $[H_2PO_4^-]$ (Fig. 3), while the correlations with the increase in intramuscular $[H^+]$ and $[P_i]$ tended to be significantly related, with *P* values of 0.06.

ATP synthesis rates and ATP cost of contraction. ATP synthesis rates and ATP cost of contraction data are presented in Fig. 4. ATP_{CK}, ATP_{GLY}, ATP_{ANA}, and ATP_{TOTAL} significantly decreased throughout the exercise test. ATP_{OX} remained within the V_{max} 95% CI (30.8 ± 8.7 mM·min⁻¹·N⁻¹, 95% CI: 23.5–38.1 mM·min⁻¹·N⁻¹) throughout the entire protocol. There was a significant, but transient, increase in ATP_{OX} during the protocol, with a small significant difference between the second and fifth minutes. The ATP cost of contraction significantly increased as a function of time during the exercise. The change in ATP cost of contraction was not significantly correlated with changes in any of the neuromuscular function measures. Data for the percent contribution of ATP_{OX} and ATP_{ANA} to ATP_{TOTAL} are presented in Fig. 5. The percentage of ATP_{TOTAL} arising from ATP_{ANA} significantly decreased over the first 2 min, while the percentage arising from ATP_{OX} significantly increased over the first 2 min, with no significant changes in either variable thereafter. The percent contribution from ATP_{OX} was significantly greater than ATP_{ANA} for minutes 2–5 (Fig. 5). During exercise at CF (i.e., the final 30 s), the percentage of ATP_{TOTAL} arising from ATP_{OX} ($83 \pm 13\%$) was significantly greater than ATP_{ANA} ($17 \pm 13\%$). The ATP_{OX} gain significantly increased throughout the exercise, while the ATP_{ANA} gain significantly decreased from the first minute to the second minute and did not significantly change thereafter (Fig. 6).

DISCUSSION

This study utilized ³¹P-MRS to quantitatively characterize skeletal muscle bioenergetics during all-out exercise. Consistent with our first hypothesis, ATP_{OX} attained and remained at peak values, while force production progressively decreased throughout the all-out exercise. As a result, the ATP_{OX} gain progressively increased in a “slow component-like” manner, similar to the $\dot{V}O_2$ slow component documented during all-out cycling exercise. Furthermore, the ATP cost of muscle contraction progressively increased throughout the all-out exercise, which is consistent with our second hypothesis. This provides evidence of an increased phosphate cost of force generation across time with such all-out exercise. In agreement

Table 1. Changes in neuromuscular function with an all-out intermittent isometric single-leg knee-extensor exercise test

	Preexercise	Postexercise	%Δ Preexercise to Postexercise
Q _{tw} , N	187 ± 26	89 ± 46†	-52 ± 26
VA, %	91 ± 5	80 ± 12†	-12 ± 11
CT _{Qtw} , ms	79 ± 9	69 ± 8†	-13 ± 28
0.5RT _{Qtw} , ms	54 ± 7	49 ± 22	-12 ± 6

Values are means ± SD. Q_{tw}, potentiated quadriceps twitch force; VA, voluntary activation; CT_{Qtw}, contraction time; 0.5RT_{Qtw}, half-relaxation time. †Significantly different from preexercise.

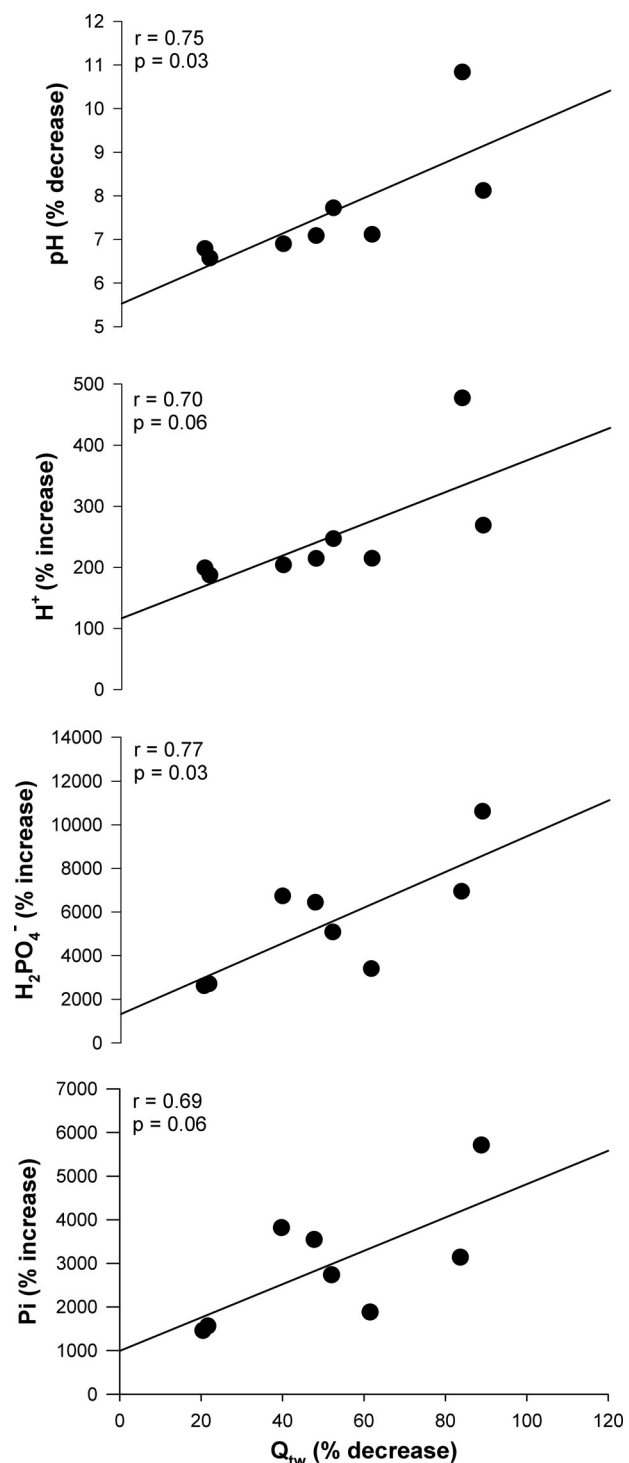


Fig. 3. Relationship between quadriceps fatigue and intramuscular metabolites. Data are expressed as % difference from baseline to end exercise for intramuscular pH and hydrogen ion ($[H^+]$), inorganic phosphate ($[P_i]$), and diprotonated phosphate ($[H_2PO_4^-]$) concentrations vs. preexercise-to-postexercise % difference for the potentiated quadriceps twitch force (Q_{tw}).

with our third hypothesis, the exercise-induced reduction in Q_{tw} was related to the changes in intramuscular $[H_2PO_4^-]$ and pH. This finding reinforces the link between the magnitude of intramuscular perturbation and peripheral fatigue that has previously been documented for large-muscle-mass cycling exer-

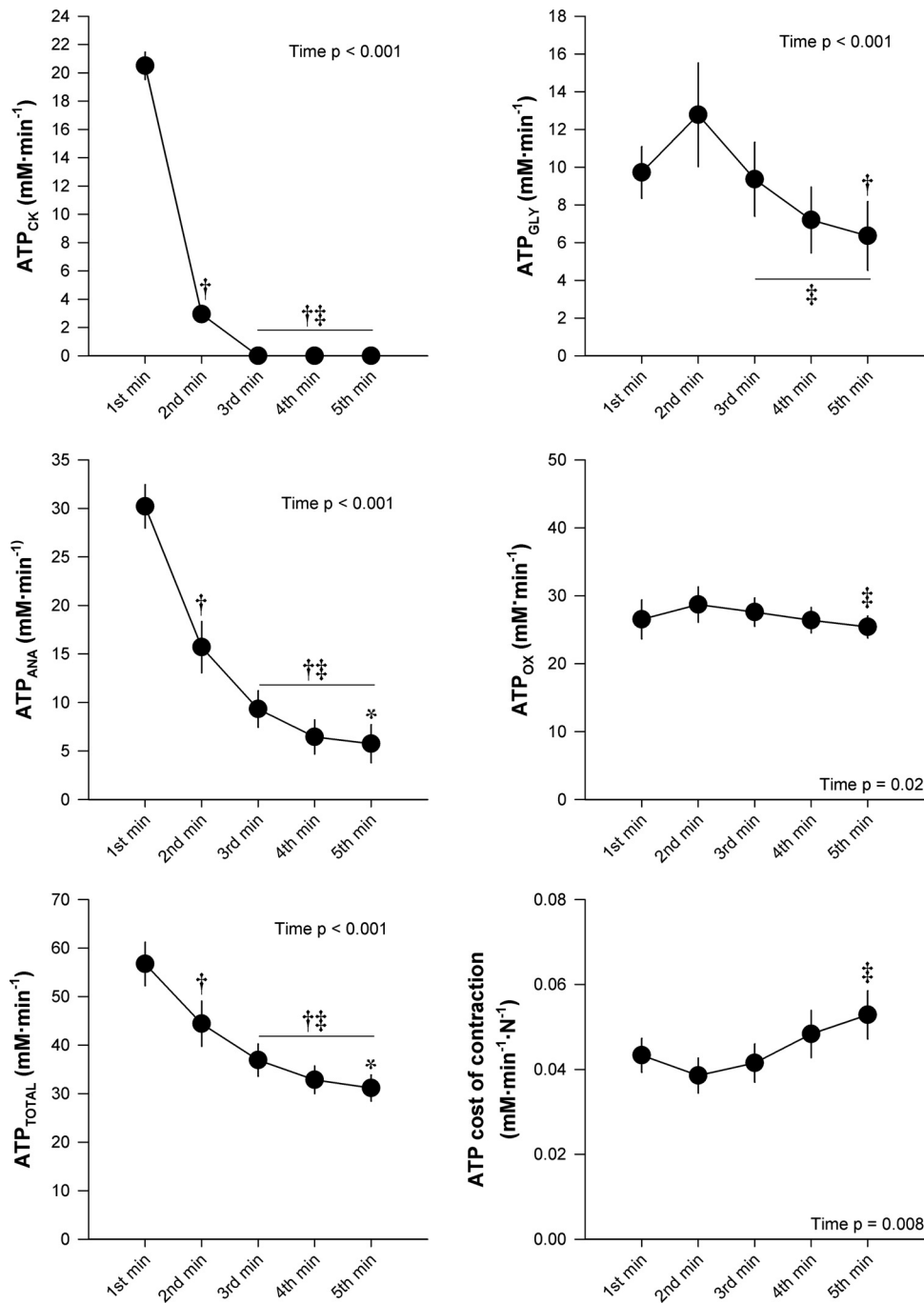


Fig. 4. Adenosine triphosphate (ATP) synthesis rates and ATP cost of contraction during 5-min all-out intermittent isometric single-leg knee-extensor protocol. Rate of ATP synthesis through creatine kinase reaction (ATP_{CK}), anaerobic glycolysis (ATP_{GLY}), cumulative anaerobic metabolism (ATP_{ANA}), and oxidative phosphorylation (ATP_{OX}), total ATPase rate (ATP_{TOTAL}), and ATP cost of contraction were determined for each minute of exercise. Significantly different: †from 1st minute, ‡from 2nd minute, *from 3rd minute ($P < 0.05$).

cise. In combination, the findings of this study offer a novel, in-depth assessment of skeletal muscle bioenergetics during all-out exercise, while also providing mechanistic insight in the determinants of the $\dot{V}O_2$ slow component and neuromuscular fatigue during exercise.

Bioenergetics of all-out exercise. The force profile and intramuscular metabolic perturbation (both magnitude and time course) in this study were similar to previous reports for both the 5-min all-out knee-extensor and 3-min all-out cycling protocols (6, 8, 52, 53). However, of importance, the present work builds upon and extends the findings of these previous studies by assessing the sources and rates of ATP synthesis

throughout the all-out exercise (Fig. 4). Consistent with the role of PCr as an energy buffer, ATP_{CK} peaked during the first minute of exercise and then rapidly decreased to no longer measurably contribute to ATP_{TOTAL} over the third through fifth minutes of exercise. ATP_{GLY} peaked during the second minute and then decreased, yet still contributed ~20% of the ATP_{TOTAL} during the final 3 min of the exercise. Concomitantly, ATP_{OX} attained maximal values (i.e., not different from V_{max}) during the first minute of exercise, which were maintained throughout the remainder of the protocol. In concert, the time course changes in ATP_{ANA} and ATP_{OX} resulted in a peak in ATP_{TOTAL} during the first minute of exercise, which pro-

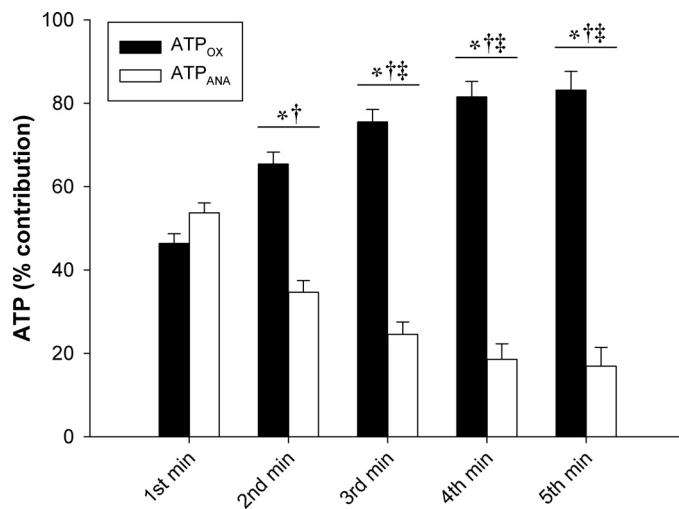


Fig. 5. Contribution of oxidative and anaerobic adenosine triphosphate (ATP) production during 5-min all-out intermittent isometric single-leg knee-extension protocol. Rates of ATP synthesis through oxidative phosphorylation (ATP_{OX}) and anaerobic metabolism (ATP_{ANA}) are expressed relative to total ATPase rate (ATP_{TOTAL}). Significantly different: *between conditions, †from 1st minute, ‡from 2nd minute ($P < 0.05$).

gressively decreased thereafter. This is consistent with the suggestion that, during fatiguing exercise, muscle force production declines so that intramuscular [ATP] is maintained within a certain range (21). Additionally, the proportion of ATP_{TOTAL} arising from ATP_{ANA} and ATP_{OX} progressively shifted from ~50/50% during the first minute to ~20/80% over the final 3 min of exercise (Fig. 5). These findings provide a novel, comprehensive characterization of skeletal muscle bioenergetics during all-out exercise.

ATP cost of contraction during all-out exercise. Importantly, normalizing ATP_{TOTAL} to force production unveiled a progressive increase in the ATP cost of contraction from the second minute of exercise onward (Fig. 4). For this exercise paradigm, the increased ATP cost of contraction is likely not related to a progressive recruitment of higher-order muscle fibers, as all motor units appear to be recruited at the onset of such all-out exercise (6, 36, 49, 56). Moreover, motor unit recruitment, assessed by electromyography, has been documented to progressively decrease during all-out cycling and knee-extension exercise (6, 56). The findings of the present study also demonstrate that the increased ATP cost of contraction is likely not related to changes in the contractile properties of the muscle fibers, as the change in ATP cost was not related to changes in neuromuscular function (i.e., Q_{tw} , $CT_{Q_{tw}}$, and $0.5RT_{Q_{tw}}$). Rather, the dramatic decline in MVC force (~63%) in the face of a moderate decrease in ATP synthesis (~40%) lends support to the hypothesis of persistent metabolism in muscle fibers that are contributing less to the overall force production as a result of fatigue. In this regard, it is interesting to note that the ATP_{ANA} gain attained a plateau after 3 min of exercise, while the ATP_{OX} gain increased continuously throughout the exercise (Fig. 6). Although it is not possible to definitively determine the metabolic contribution of motor units comprising different muscle fiber types with the present experimental design, these results are consistent with findings in isolated single myocytes that fatigue-resistant muscle fibers exhibit a greater oxidative ATP cost of contraction during fatiguing

exercise, whereas more fatigable muscle fibers reduce metabolic demand proportionally to the fall in tension (17).

Peripheral fatigue and intramuscular metabolic perturbation during all-out exercise. The all-out exercise-induced central and peripheral fatigue and intramuscular metabolic perturbation were closely related in the present study (Fig. 2). Specifically, the reduction in Q_{tw} was correlated to changes in $[H_2PO_4^-]$, $[H^+]$, $[P_i]$, and pH. These findings are consistent with a previous report from our group that documented similar relationships during high-intensity cycling exercise (2). Collectively, these findings demonstrate that the link between exercise-induced peripheral fatigue and intramuscular metabolic perturbation is robust across exercise modalities involving quite different amounts of muscle. Mechanistically, the robustness of these relationships provides further, in vivo, evidence of an important role of these metabolites in the development of peripheral fatigue (1). Interestingly, while a low cellular pH or elevated $[P_i]$ has been demonstrated to diminish skeletal muscle function (13, 30), it has recently been demonstrated that these metabolites can also interact synergis-

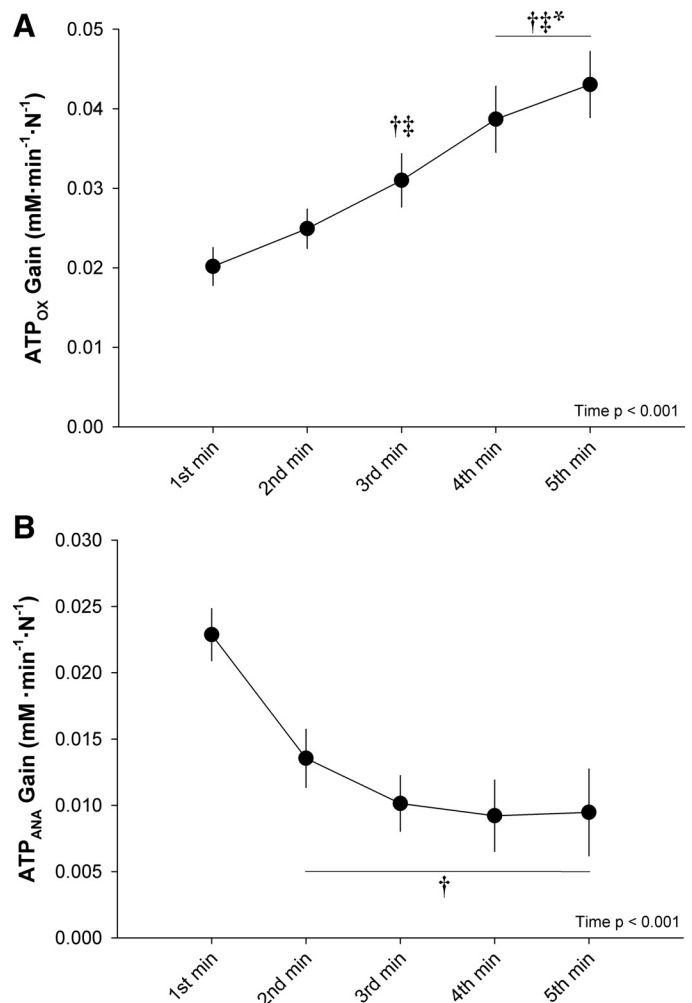


Fig. 6. Oxidative and anaerobic adenosine triphosphate (ATP) gain during 5-min all-out intermittent isometric single-leg knee-extension protocol. Gains were determined as the rate of ATP synthesis through oxidative phosphorylation (ATP_{OX}) and anaerobic metabolism (ATP_{ANA}) normalized to the integrated force. Significantly different: †from 1st minute, ‡from 2nd minute, *from 3rd minute ($P < 0.05$).

tically in the development of peripheral fatigue (41). Moreover, the apparent differences in the time course changes between the intramuscular metabolites and force production in the present study further suggest that fatigue development is likely a complex process with synergistic mechanisms.

Implications for $\dot{V}O_2$ slow component. The present data suggest that an intramuscular " $\dot{V}O_{2\max}$ " was attained throughout the all-out exercise, as ATP_{OX} values were similar to the peak rate of in vivo oxidative ATP synthesis (i.e., $V_{\max}$). This finding is consistent with the attainment of pulmonary $\dot{V}O_{2\max}$ during whole body all-out exercise protocols (7, 9, 22, 52, 53). However, such findings, at the muscle level, bolster the use of the all-out exercise protocol as a simple and practical test to determine CF and muscle aerobic capacity in a single test. Interestingly, ATP_{OX} maintained maximal values despite a ~63% reduction in force production in the present study, resulting in the progressive increase in ATP_{OX} gain throughout the exercise (Fig. 6). Importantly, this reveals an intramuscular slow component in the rate of oxidative phosphorylation elicited during the all-out exercise that arose from a progressive loss of muscle efficiency within the exercising skeletal muscle. These findings further support that the $\dot{V}O_2$ slow component arises from the exercising skeletal muscle (45, 47) and that progressive recruitment of higher-order muscle fibers is not requisite to evoke the $\dot{V}O_2$ slow component (18, 52, 56, 59). These findings are also consistent with the expression of a pulmonary $\dot{V}O_2$ slow component during a 3-min all-out cycling test (52, 56).

Implications for CF. From the outset of discussing the implications for CF, although it is recognized that CP and CF are different, as the implications of assessing both CP and CF are very similar, for clarity, only the term CF will be used here. The growing body of evidence supports that CF is predominantly determined by oxidative energy production (3, 5, 12, 14, 26, 37, 38, 46, 52, 54). This interpretation comes from assessing the alteration in CF as a result of manipulations in oxygen delivery. However, the intramuscular contribution of ATP_{OX} and ATP_{ANA} to ATP_{TOTAL} has not been quantified during exercise at CF. In this study, it was demonstrated that ATP_{OX} and ATP_{ANA} contributed ~80% and ~20% to ATP_{TOTAL} during exercise at CF (i.e., final 30 s), respectively (Fig. 5). These findings support the concept that CF is predominantly determined by oxidative energy production and explains why manipulating oxygen delivery has such a significant impact on CF. Importantly, however, the findings of this study demonstrate that CF is not fully independent from ATP_{ANA} (specifically ATP_{GLY}), which contributed ~20% to ATP_{TOTAL} . In light of the observation that intramuscular metabolites do not progressively change during exercise at or slightly below CF (26, 46, 52), this ~20% ATP_{ANA} contribution is likely such that energy utilization and replenishment are in equilibrium and, therefore, no net change occurs. In support of this interpretation, pH was constant across the latter portion of the exercise protocol in the present study, despite the continued ~20% contribution from ATP_{ANA} .

Conclusions. The quantitative characterization of skeletal muscle bioenergetics during all-out exercise revealed that within this paradigm a small muscle mass exhibits an intramuscular $\dot{V}O_2$ slow component, as well as a progressive increase in the phosphate cost of force generation. Additionally, the intramuscular metabolic perturbation was closely related to

the development of peripheral fatigue during the all-out exercise. Collectively, these findings provide direct evidence supporting the concept that the $\dot{V}O_2$ slow component arises from within the active skeletal muscle and that the force generating capacity of skeletal muscle is closely linked to the intramuscular metabolic milieu.

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DISCLOSURES

No conflicts of interest, financial or otherwise, are declared by the authors.

AUTHOR CONTRIBUTIONS

R.M.B., G.L., T.J.H., M.A., and R.S.R. conceived and designed research; R.M.B., G.L., T.J.H., M.A., and R.S.R. performed experiments; R.M.B., G.L., T.J.H., M.A., and R.S.R. analyzed data; R.M.B., G.L., T.J.H., M.A., and R.S.R. interpreted results of experiments; R.M.B., G.L., T.J.H., M.A., and R.S.R. prepared figures; R.M.B., G.L., T.J.H., M.A., and R.S.R. drafted manuscript; R.M.B., G.L., T.J.H., M.A., and R.S.R. edited and revised manuscript; R.M.B., G.L., T.J.H., M.A., and R.S.R. approved final version of manuscript.

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